Kyle Medina

San Diego, CA | (619) 486-4123 | medinakyle2@gmail.com | U.S. Citizen
linkedin.com/in/kyle-medina-aerospace-engineer | kylemedina-aerospace.github.io

EDUCATION

University of California, Los Angeles - *B.S., Aerospace Engineering | Expected June 2027 | GPA: 3.28* **Los Angeles, CA**

• **Relevant Coursework:** Rocket Propulsion Systems; Aerodynamics; Intermediate Fluid Dynamics; Thermodynamics; Transport Phenomena; Dynamic Systems Control; Modeling & Analysis of Dynamic Systems; Space Technology; Dynamics of Particles & Rigid Bodies; Statics & Strengths of Materials; Science of Engineering Materials; Electrical & Electronic Circuits

In Progress: Aircraft Propulsion Systems; Analysis of Aerospace Structures; Flight Mechanics, Stability, & Control of Aircraft

Lund University - *Exchange Semester, Aerospace Engineering | August 2025 - December 2025* **Lund, Sweden**

ENGINEERING PROJECTS & EXPERIENCE

Rocket Project at UCLA - *Test Engineer, Engine Team | Sept 2024 - June 2026* **Los Angeles, CA**

• Assembled the engine static fire test stand, contributed to preliminary design, and installed it in the field with the test crew for hot-fire testing of the Prometheus engine

• Conducted injector water-flow tests (setup, test operations, data recording) to characterize flow behavior and pressure drop, and built ignition test hardware

Engineering Design: Airplanes - *Team Leader, 5-Person Team | April 2024 - June 2024* **Los Angeles, CA**

• Led full-cycle design and build of an RC aircraft, using CAD, MATLAB, and XFLR5 for aerodynamic modeling and control

• Completed a successful self-propelled takeoff, 2.5-minute flight with two barrel rolls, 80+ ft/s top speed, and safe landing

Independent Projects **San Diego, CA**

Motorcycle Build - Refurbished Bicycles - Designed & Built Custom Furniture - Rebuilt Suspension & Brakes on Vehicles

RESEARCH EXPERIENCE

Thermal Science Laboratory for Semiconductor Packaging & Integration, UCLA **Los Angeles, CA**

Undergraduate Student Researcher | June 2026 - Present

• Lead development of a scalable, batch-capable process to bond PTFE/PP membranes to 3D-printed resin manifolds for a two-phase electronics cooling device, replacing a slow manual adhesive process that left gaps

• Wrote bonding requirements from operating conditions (110 °C hot-wet service, leak-tight at 72 kPa, adhesive-free active area) and derived process pressure limits from material datasheets

• Ran a trade study of four adhesive families and application methods (stencil, die-cut) and down-selected two candidates; presented findings to the lab group and begin bonding and leak-testing prototypes in fall 2026

TEMP Lab & Cai Group at UC San Diego - *Undergraduate Student Researcher | Summers 2025 & 2026* **La Jolla, CA**

Membrane permeability verification study (June 2026 - September 2026)

• Tested whether permeability of copper mesh, Cu-coated glass fiber, and quartz membranes in series matched series-resistance predictions across 120 flow trials, using Darcy's law and Laplace capillary-pressure prediction

• Built a MATLAB analysis pipeline with Monte Carlo uncertainty propagation (5,000 resampled fits per sample) and automated validity checks on Darcy linearity, zero-pressure intercept, and pore Reynolds number

• Found measured and predicted values disagreed and traced the likely cause to un-modeled test-rig flow resistance, not sample properties (~0.5 m unaccounted head across samples); presented the diagnosis and a blank-run calibration plan to the PI

Hydrogel membrane fabrication (June 2025 - August 2025)

• Developed the repeatable method used by the lab to bond uniform ~25 µm hydrogel films to porous glass fiber substrates without fiber infiltration (layered tape borders, doctor blade), part of a group effort demonstrating heat flux above 800 W/cm²

• Designed a microscopy-based technique to quantify coating thickness and uniformity using an in-frame razor blade as a pixel-to-distance reference, providing the validation method for the fabrication process

LEADERSHIP & ACTIVITIES

Engineering Ambassadors - Ambassador | April 2024 - Present **Los Angeles, CA**

• Represent UCLA Engineering during 35+ tours for prospective students and alumni, as well as 1-on-1 prospective student Q&As, 6 Engineering Commencement Ceremonies, and 3 Bruin/Engineering Welcome Days

• Deliver engineering presentations at Santa Monica College and collaborate with faculty and donors to promote STEM engagement in the local community and on campus

SKILLS

• **Software & Tools:** Python, MATLAB, CAD/SolidWorks, XFLR5, Microsoft Office, Google Workspace

• **Fabrication & Lab:** Lathe, Mill, Drill Press, Band Saw, 3D Printing (filament & resin), Laser Cutting, Microscope, Micropipette, Flow Loop & Permeability Testing

• **Competencies:** Requirements Definition, Trade Studies, Test Planning, Root Cause Analysis, Leadership, Communication

PROFESSIONAL EXPERIENCE

San Diego Rock Supply - *Operator/Trainer/Cashier | May 2023 - September 2023* **El Cajon, CA**

• Operated a CAT 906 front loader; trained new hires; oversaw deliveries, machinery maintenance, & equipment lock-up

Vons - *Courtesy Clerk | July 2024 - September 2024* **Lake Murray, CA**

• Maintained store cleanliness and organization; supported front-end operations; provided excellent customer service